

Trilateration, 3 beacons

$$B_1 = (10, 1) \quad B_2 = (2, 6) \quad B_3 = (1, 1)$$

true location of receiver $\underline{u} = (3, 3)$

true range:

$$R_{B_1} = \sqrt{\quad} = 7.28$$

$$R_{B_2} = \sqrt{\quad} = 5.83$$

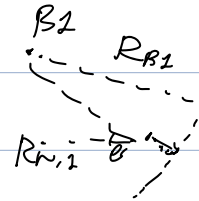
$$R_{B_3} = \sqrt{\quad} = 2.83$$

initial guess $\underline{u}_N = (12, 8)$

$$R_{N,1} = \sqrt{\quad} = 7.28$$

$$R_{N,2} = \sqrt{\quad} = 4.47$$

$$R_{N,3} = \sqrt{\quad} = 13.04$$



The Jacobian matrix becomes:

$$H = \begin{bmatrix} \frac{12-10}{7.28} & \frac{8-1}{7.28} \\ \frac{12-2}{4.47} & \frac{8-6}{4.47} \\ \frac{12-1}{13.04} & \frac{8-1}{13.04} \end{bmatrix} = \begin{bmatrix} 0.275 & 0.962 \\ 0.895 & 0.447 \\ 0.844 & 0.537 \end{bmatrix}$$

From $H \delta \underline{u} = \delta \underline{r}$, Moore-Penrose
pseudo-inverse

$$\Rightarrow H^T H \delta \underline{u} = H^T \delta \underline{r}$$

$$\Rightarrow (H^T H)^{-1} (H^T H) \delta \underline{u} = (H^T H)^{-1} H^T \delta \underline{r}$$

$$\Rightarrow \delta \underline{u} = (H^T H)^{-1} H^T \delta \underline{r}$$

$$= \begin{bmatrix} 1.418 & -1.121 \\ -1.121 & 1.594 \end{bmatrix} H^T \delta \underline{r}$$

$$\delta \underline{r} = \begin{bmatrix} 0 \\ 1.36 \\ -1.44 \end{bmatrix}$$

$$z = \begin{bmatrix} -5.029 \\ 0.525 \end{bmatrix}$$

then $\underline{u}_1 = \begin{bmatrix} 6.971 \\ 8.515 \end{bmatrix}$

$$\|8u\| = 5.056 > \gamma.$$

continue iteration. then